UNIT III — FEATURE DETECTION, MATCHING & IMAGE SEGMENTATION | Quick-Revision Sheet

01 · RANSAC

Random Sample Consensus

DEFINITION

An iterative method to estimate a mathematical model from data containing outliers, by identifying & excluding them.

THEORY

- Works by finding outliers & fitting model only to inliers
- N samples needed to fit a model (e.g. 2 points → a line)
- Even ONE sample brings information → votes in model space
- Collect votes from all samples → seek consensus

PROBLEM IT SOLVES

Fitting y=mx+c (or F-matrix, homography) to noisy data where least-squares fails due to outliers.

PIPELINE

Random subset → Fit model → Count inliers → Repeat N iters → Best model

STAGES

- Randomly select subset of dataset
- Fit model to selected subset
- Determine no. of outliers (points off model)
- Repeat steps for a prescribed no. of iterations

```
Line fit: y = mx + c (red)  |  blue = raw data with outliers
```

USES

- Fundamental matrix (stereo vision)
- Feature-based object detection
- Video stabilization (frame registration)

LIMITATIONS

- No guaranteed iteration bound for success
- Fails with very high outlier ratio
- Needs careful threshold tuning; non-deterministic

02 · CORNERS

Corner Detection (Moravec & Harris)

DEFINITION

A corner is an interest point of high boundary curvature where intensity varies strongly in ALL directions - a stable, uniquely-matchable feature.

THEORY

- Flat region: no variation in any direction (bad feature)
- Edge: variation in only 1 direction (still a bad feature)
- Corner: variation in ALL directions (great feature!)
- Corners = uniquely identifiable + stable (little motion → big change)

PROBLEM IT SOLVES

Finding trackable, matchable points across frames/views (stereo correspondence, tracking) where plain patches fail on repetitive/flat regions.

PIPELINE

Place window → Shift in all dir. → Auto-correlate → Score R → Threshold + NMS

STAGES (HARRIS)

- $E(u,v) = \Sigma\, w(x,y)[I(x+u,y+v) - I(x,y)]^2$
- Taylor expand: $I(x+u,y+v) \approx I + uI_x + vI_y$
- Reduce to matrix form using structure matrix M
- Compute score R from eigenvalues of M → threshold + NMS

```
M = Σw(x,y)·[[Ix², IxIy],[IxIy, Iy²]]   R = det(M) - k·trace(M)²
```

ADVANTAGES

- Harris: fast, rotation & illumination robust
- Popular, simple to compute (OpenCV: Shi-Tomasi variant)

LIMITATIONS

- Moravec: not rotation invariant, noise-sensitive
- Harris: not scale invariant

03 · EDGES

LoG & Difference of Gaussian (DoG)

DEFINITION

DoG = difference of two Gaussian-smoothed versions of an image at different σ; a fast approximation of the Laplacian of Gaussian (LoG) used to detect edges.

THEORY

- LoG: smooth with Gaussian, then apply Laplacian → zero crossings = edges
- DoG curve shape ≈ LoG curve shape (verified via plots)
- Subtracting two blurred images cancels low-freq content, keeps edge band

PROBLEM IT SOLVES

Detecting intensity discontinuities (edges) robustly while suppressing noise; avoids costly explicit Laplacian computation.

PIPELINE

Blur σ_1 → g_1 → Blur σ_2 → g_2 → $g_1 - g_2$ = DoG → Zero-crossing → Threshold → edges

STAGES

- Convolve f(x,y) with $G_{\sigma 1} \rightarrow g_1(x,y)$
- Convolve f(x,y) with $G_{\sigma 2} \rightarrow g_2(x,y)$
- DoG = $g_1 - g_2 = (G_{\sigma 1} - G_{\sigma 2}) * f(x,y)$
- Find zero crossings; threshold weak ones caused by noise

```
Gσ(x,y) = 1/√(2πσ²) · exp(-(x²+y²)/2σ²)   DoG ≙ Gσ1 - Gσ2
```

ADVANTAGES

- Cheap approximation of LoG (no 2nd derivative needed)
- Basis of SIFT's scale-space extrema detection

LIMITATIONS

- σ1, σ2 choice affects quality
- Zero-crossings sensitive to noise, need thresholding

04 · HOUGH

Hough Transform

DEFINITION

A voting-based feature extraction technique that maps image-space points into a parameter space to detect regular curves (lines, circles, ellipses).

THEORY

- Each point in parameter (m,c) space represents a candidate line/model
- Cartesian y=mx+c is problematic for vertical lines (m unbounded)
- Solved via polar parametrization → bounded ρ, θ
- Generalized HT: works for shapes without simple analytic form

PROBLEM IT SOLVES

Grouping scattered/gapped/noisy edge points into global lines or curves via consensus voting.

PIPELINE

Edge image → Plot curve per point → Accumulate votes → Find peaks → Map back → lines

STAGES

- Rearrange y=mx+c → c=(-x)m+y for each point (x_i, y_i)
- Or use ρ = x cosθ + y sinθ (polar form)
- Each edge pixel votes in accumulator over (ρ,θ) or (m,c)
- Peaks in accumulator = detected lines

```
ρ = x cosθ + y sinθ  (ρ≥0, 0°≤θ≤360°)  |  ρ=dist. from origin, θ=normal angle
```

ADVANTAGES

- Robust to noise, gaps & partial occlusion
- Polar form bounded → handles vertical lines

LIMITATIONS

- Computationally expensive (large accumulator space)
- Largely superseded by learning-based methods

05 · SIFT

Scale-Invariant Feature Transform

DEFINITION

A keypoint detector + descriptor that finds local features invariant to scale, rotation & (partly) illumination, using DoG scale-space extrema.

THEORY

- Build Gaussian scale-space across octaves (increasing σ)
- Take DoG between adjacent scales → locate extrema
- Assign dominant orientation → rotation invariance
- Build 128-D descriptor from local gradients

PROBLEM IT SOLVES

Matching features reliably across images with scale/rotation/viewpoint/illumination changes (stereo, panorama, recognition).

PIPELINE

Scale space → DoG → Locate extrema → Filter edge/ low-contrast → Orientation → Descriptor

STAGES

- Construct scale space: G(σ)*I, G(kσ)*I … per octave
- DoG extrema → candidate keypoints (compare 26 neighbours)
- Sub-pixel localize; reject low-contrast/edge responses
- 36-bin gradient-direction histogram (10°/bin) → orientation
- 16×16 window → 4×4 cells × 8 orientation bins = 128-D

```
Keypoint = local extremum of D(x,y,σ) in scale-space;
descriptor = concatenated gradient histograms
```

ADVANTAGES

- Scale + rotation invariant; highly distinctive
- Robust to illumination & viewpoint change

LIMITATIONS

- Computationally heavy; high-dim (128-D) descriptor
- Slower than SURF; historically patented

06 · SURF/GLOH

SURF & GLOH Descriptors

DEFINITION

SURF (Speeded-Up Robust Features): fast SIFT approximation using box filters. GLOH (Gradient Location-Orientation Histogram): SIFT variant with log-polar binning.

THEORY

- SURF: box filters approximate Laplacian; Haar wavelets give orientation
- SURF ~3× faster than SIFT; good for blur & rotation, weak on illumination/viewpoint
- GLOH: 17 spatial (log-polar) × 16 orientation bins = 272-D histogram
- GLOH uses PCA to reduce 272-D → 128-D (efficient)

PROBLEM IT SOLVES

SURF: real-time-capable distinctive features. GLOH: more discriminative/robust descriptor than plain SIFT.

PIPELINE

Integral image → Box filter (L_{xx}, L_{yy}, L_{xy}) → Hessian extrema → Haar orient. → Descriptor

STAGES

- **SURF:** box filter approx. of L_{yy}, L_{xy} → approx. Hessian det. → blob keypoints → Haar-wavelet orientation & descriptor
- **GLOH:** gradients in circular region → log-polar bin (17×16) → 272-D histogram → PCA → 128-D

```
Box filter ≈ Gaussian 2nd derivative (e.g. Lyy: [1,-2,1] pattern)  |  GLOH: 272-D → PCA → 128-D
```

ADVANTAGES

- SURF: fast (real-time), blur/rotation robust
- GLOH: more discriminative than SIFT

LIMITATIONS

- SURF: weak under illumination/viewpoint change
- GLOH: costlier (272-D) before PCA reduction

07 · SEGMENTATION

Image Segmentation & Gestalt Theory

DEFINITION

Segmentation divides an image into regions using similarity or discontinuity principles; Gestalt theory explains the human-perception grouping cues behind it.

THEORY

- Gestalt: "the whole is greater than the sum of its parts"
- Grouping factors: proximity, similarity, parallelism, symmetry
- Also: common fate, common region, continuity, closure
- Illustrated by Muller-Lyer illusion (perception ≠ reality)

PROBLEM IT SOLVES

How to group pixels/elements into perceptually meaningful regions/objects - foundation for region-based segmentation algorithms.

PIPELINE

Raw pixels → Apply grouping cue → Group elements → Perceived regions

STAGES

- Identify grouping factor present (proximity/similarity/colour/motion)
- Group elements sharing that factor
- Combine multiple cues if needed
- Result: organized, perceptually meaningful regions

```
No single formal equation - qualitative rules of perceptual organization guide clustering/region segmentation
```

ADVANTAGES

- Explains human-perceived grouping
- Inspires similarity/proximity-based segmentation

LIMITATIONS

- Fairly descriptive; rules insufficiently defined
- Not directly algorithmic / hard to formalize

08 · REGION-BASED

Region Growing & Split-and-Merge

DEFINITION

Region Growing merges pixels into larger homogeneous regions from seeds. Split-and-Merge recursively subdivides (quadtree) then merges regions by homogeneity.

THEORY

- Region growing: principle of similarity; homogeneity criteria = gray level, colour, texture, shape
- Split-Merge = "Divide and Conquer" / Top-Down approach
- Phase I Split, Phase II Merge (two distinct phases)

PROBLEM IT SOLVES

Partitioning an image into homogeneous regions without a single global threshold; adapts to local similarity.

PIPELINE

Whole img = 1 region → Test homogeneity → Split into 4 quadrants → Merge similar adj. → Final regions

STAGES

- **Region grow:** select seed → grow using $|f(x,y) - f(x',y')| \le T$ → stop when no pixel qualifies
- **Split:** test homogeneity → split into 4 if not met → repeat till stop criterion
- **Merge:** merge adjacent regions sharing common property → repeat till no more merges

```
Homogeneity test: |f(x,y) - f(x',y')| ≤ T  (Threshold, e.g. T=4)  |  Quadtree = R1,R2,R3,R4...
```

ADVANTAGES

- Region grow: simple for well-defined seeds
- Split-Merge: adapts to arbitrary shapes/sizes

LIMITATIONS

- Region grow: seed & noise sensitive, may over-grow
- Split-Merge: blocky quadtree artifacts